Supplemental Material for "Bitwise-Reproducible Reduction Across Cluster Reshape: A Compiler/Fabric ABI"

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This companion carries three appendices to the manuscript: the fifteen assumptions on which Claim 1 and Property 1 rest (§A); the assignment-algorithm catalogue and the instance-suite coverage table (§B); and the reproducibility record, including what each per-run verdict is and the standing disclosure that the software oracle is not independent of the RTL (§C). Section, table, claim, definition and equation numbers that are not defined here are the manuscript's own, resolved from its auxiliary file rather than retyped. Every number is generated from the same data files the manuscript uses, so the two documents cannot disagree.

Appendix A
The Fifteen Assumptions

Five compiler obligations, discharged by the compiler or the user; ten fabric invariants, established by RTL inspection. The four marked † are hypotheses of Property 1 only: violating one can stop a run but cannot alter a root the fabric does produce.

Compiler obligations. C-sw (single-writer allocation for remote destinations): partitioning each PE's DMEM into *receive buffers* — addresses named as `target_addr` by a remote `SEND` — and *local temporaries*, the image contains for every PE p and receive buffer $a \neq 0$ exactly one write to (p, a), and it is remote. Local temporaries are exempt: the allocator reuses them and their safety comes from in-order issue rather than single assignment. The scoping is load-bearing and the two classes are disjoint by construction; on the emitted images of 562 configurations the recycled and remotely-written address sets never intersect, and no operand is read after the write that recycles its slot (zero such events in 397,422). C-bind and C-leaf are Defs. 4 and 5, checked as post-conditions on the emitted image. C-mem (memory feasibility): every PE names at most $\texttt{DMEM\_DEPTH} = 512$ distinct DMEM addresses and every emitted router id is below the adjacency sentinel (12 bits, at most 4,094). Neither is enforced at run time — the fabric truncates DMEM and `SEND` target addresses to ten bits, so an over-budget image aliases two addresses onto one slot silently, and out-of-range router ids alias onto low ids, which is what stopped one instance from ever settling (§XI, L0) — but both are cheap to discharge, peak occupancy being an $O(N)$ read-off the allocator already maintains. † C-route: the emitted tables encode destination-correct, loop-free paths. We claim those two properties only; the tables are minimal-distance rather than dimension-ordered, so we do not claim an acyclic channel-dependency graph.

Fabric invariants. F-sb-mono: for every receive buffer a, `scoreboard[a]` transitions $0 \to 1$ at most once per program, at the clock edge of the unique remote write, and stays set until reset. F-sb-gate: `COMPUTE` stalls DE→EX until both operand bits are 1, `SEND` until its source bit is. F-fw: a DMEM read at cycle t returns the value committed by the unique write at cycle $\leq t$, even across the adjacent-cycle race. F-inorder: each PE issues in order through IF→DE→EX, one instruction at a time, sampling both operands at the DE read port strictly before writing back in EX — which is what makes local-temporary reuse safe, since the scoreboard cannot, a recycled slot's bit being already set. F-gci-haz: from `gci_start` until the GCI agent commits, DE stalls any `COMPUTE` or `SEND` naming the pending destination. F-arb: strict priority PIU $>$ GCI $>$ FSM with a GCI retry latch; no write is silently dropped. F-fadd: `fp64_add.sv` is combinational and stateless, its output a pure function of the two 64-bit inputs sampled at the EX clock edge. † F-credit: per link and interval, #credit_returns $\leq$ #buffer_reads with bounded wait. † F-misroute: packets whose `dest_gpu_id` does not match the local PE are dropped, not silently consumed. † F-watchdog: a stall exceeding $\texttt{WATCHDOG\_MAX} = 20{,}000$ cycles raises `P_ERROR`, as does an invalid opcode immediately.

Appendix B
Assignment Algorithms and Suite Coverage

Table I lists the eighteen assignment algorithms, ordered by family and deliberately not by membership: below the top four they are not distinguishable from one another on a suite of this size and should be read as a block. A zero share does not mean an algorithm is useless — it means the algorithm never matched the instance minimum here, and several zero-share algorithms are close seconds. Where an algorithm is not evaluable on every instance the denominator is the eligible subset; `ExactBB` in particular is size-gated beyond $N \approx 64$, so its figure reflects unavailability rather than defeat. The mapping stage is absent by design: all ten mapping algorithms attain the champion cycle count on 27 of 39 instances, so no per-mapping share is reportable. Provenance for the

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TABLE I
ASSIGNMENT (S2) ALGORITHMS BY FAMILY, WITH CHAMPION-CLASS MEMBERSHIP ACROSS THE 39 HW-VALIDATED INSTANCES — THE SHARE OF INSTANCES ON WHICH THE ALGORITHM PRODUCES A CONFIGURATION WHOSE MEASURED CYCLE COUNT EQUALS THE INSTANCE MINIMUM. ⋆ MARKS THE FOUR MEMBERS OF THE DERIVED PORTFOLIO (§VII-F). MEMBERSHIP IS INVARIANT TO THE TIE STRUCTURE OF §VII-E; A WIN COUNT IS NOT, WHICH IS WHY THIS COLUMN REPLACES THE WINNER SHARES OF EARLIER VERSIONS.

Algorithm	Family	In champion class
`Frederickson`⋆	tree DP	12.8%
`Lukes`⋆	tree DP	48.7%
`ClHEFT`	hierarchical	0.0%
`ParSub`⋆	hierarchical	41.0%
`SubPack`	hierarchical	5.1%
`DKway`	graph partition	2.6%
`RecBisect`	graph partition	17.9%
`CPFD`	clustering	0.0%
`Chrétienne`	clustering	0.0%
`DSC`⋆	clustering	7.7%
`MLfm`	multilevel	2.6%
`Centroid`	geometric	2.6%
`LvlHLF`	structural	0.0%
`Spine`	structural	2.6%
`CPOP`	list scheduling	0.0%
`DFSseed`	list scheduling	2.6%
`HEFT`	list scheduling	0.0%
`ExactBB`	exact/B&B	2.6%

TABLE II
INSTANCE-SUITE CELL COUNTS BY N-BUCKET AND TOPOLOGY; ROWS AND COLUMNS BOTH SUM TO THE 40-INSTANCE TOTAL. 23 OF THE POPULATED CELLS HOLD TWO INSTANCES OR FEWER AND 4 ARE EMPTY, WHICH IS THE BASIS FOR L5.

N-bucket	Lin.	Ring	Mesh	Torus	Fat-tree	Hyp.	Total
8–32	1	2	1	1	1	1	7
33–256	1	1	1	3	2	1	9
257–1K	2	2	1	3	2	2	12
1K–4K	–	1	1	2	3	2	9
8K	–	–	–	1	1	1	3
Total	4	6	4	10	9	7	**40**

implementations from the literature: list scheduling follows HEFT and CPOP [1] and Hu's level-based bound [2]; tree dynamic programming follows Lukes [3]; clustering follows DSC [4]; multilevel and graph partitioning follow Karypis and Kumar [5] with Fiduccia–Mattheyses refinement [6]. The remainder are our own constructions and are marked as such in the released source. Table II gives the coverage the suite achieves; its unevenness is what bounds every scale-dependent statement in §VII-F.

APPENDIX C
REPRODUCIBILITY

Artefacts. The ten fabric sources of §VII, the compiler and its testbench generator, the mutation harness for both injected classes, the pinned-FBT implementation of §VIII and the result files behind every table and figure are released with the paper and will be deposited under a persistent identifier; the three FPGA bring-up wrappers need IP and board constraints we do not redistribute. Everything reported here was tested under Vivado xsim 2025.2 on Linux. The compiler is Python 3.9+, deterministic under insertion-ordered dict semantics, with RNG seeds pinned at 42.

What each verdict is. The per-run verdict is the emitted testbench's comparison of the fabric's result against the canonical root carried as a compile-time `EXPECTED` constant. The C-* obligations are a static audit over the released images (§V), not checks that ran inside the compile loop. We ship no SystemVerilog assertion suite and no claim here rests on one.

The oracle is not independent, and this is where that is recorded. The Python $\mathrm{fadd}_{\mathrm{rtl}}$ mirror is a transcription of `fp64_add.sv`, not an independent model: it is bit-identical to the repaired RTL on 160,800 vectors (zero mismatches), and on the as-submitted pair — the negative control — the two agree with each other while both disagree with correctly-rounded addition on 15,717 of 62,320 vectors (§IV-H, L13).

Reproducing a root. Leaf i carries L_i from (5), and every HW-validated configuration at the same N produces a 64-bit root whose hash matches the corresponding column of Table IV.

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